

Project Deliverable 3

Hypothesis & Statistical Testing

GREAT LAKES
Erie, Huron, Michigan, Ontario, Superior
PARTICULATE NUTRIENTS
Particulate Total Nitrogen (PTN), Particulate Total Phosphorus (PTP), Particulate Organic Carbon (POC), Total Suspended Solids (TSS)

QUESTION 1

How have nutrient particle levels changed over time in Lake Michigan? What predictions, or assumptions, can I make about today's levels based on past trends?

H0: Nutrient particle levels in Lake Michigan are not changing over time. $\beta_1 = 0$

H1: Nutrient particle levels in Lake Michigan are changing over time. $\beta_1 \neq 0$

To assess potential changes in particulate nutrient levels in Lake Michigan over time, I applied a linear regression model to log-transformed nutrient concentration data. I focused on PTN, PTP, POC, and TSS, primarily because these four had the fewest missing data points in the dataset.

Before interpreting the models, I checked the standard regression assumptions—linearity, constant variance, and normality—using diagnostic plots, and these all looked generally okay for the four variables. Since this is a time-series dataset, I knew I had to check for independence, so I ran autocorrelation function (ACF) plots of the residuals to ensure they didn't violate that assumption.

Looking at the ACF plots, I noticed that for PTN and POC, a few lags fell outside the 95% confidence intervals. This suggests there might be some lingering dependency between observations over time (autocorrelation), meaning I might be violating the assumption of independent residuals for these two. On the other hand, the PTP and TSS models didn't show much autocorrelation, with only a few, if any, lags breaking the bounds, so those seem okay. Overall, while most of the assumptions seem to hold, I should probably keep in mind that the independence assumption is a bit shaky for PTN and POC when interpreting the results.

OUTCOME:

PTN:

Reject H_0 : The model returned $p = 9.558e-06$, indicating there is strong evidence that PTN levels in Lake Michigan have changed over time. Kindly note that PTN shows evidence of autocorrelation in the residuals, so the exact p-value should be interpreted with some caution; however, the trend remains strongly significant.

PTP:

Fail to Reject H_0 : The model returned $p = 0.9541$, indicating there is no evidence that PTP levels in Lake Michigan change over time.

POC:

Fail to reject H_0 : The model returned $p = 0.400$, indicating there is no evidence that POC levels in Lake Michigan change over time.

TSS:

Fail to reject H_0 : The model returned p-value = 0.6388, indicating there is no evidence that TSS levels in Lake Michigan change over time.

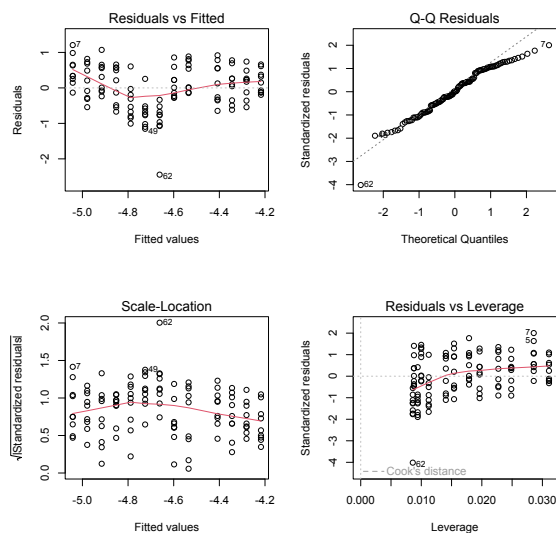
CONCLUSION:

Based on linear trend tests in Lake Michigan, three of four particulate nutrients (PTP, POC, and TSS) show no statistically significant change in measurements over time, while PTN shows strong evidence of a time-based trend. Therefore, **particulate nutrient patterns appear mostly stable across time for PTP/POC/TSS, but not for PTN.**

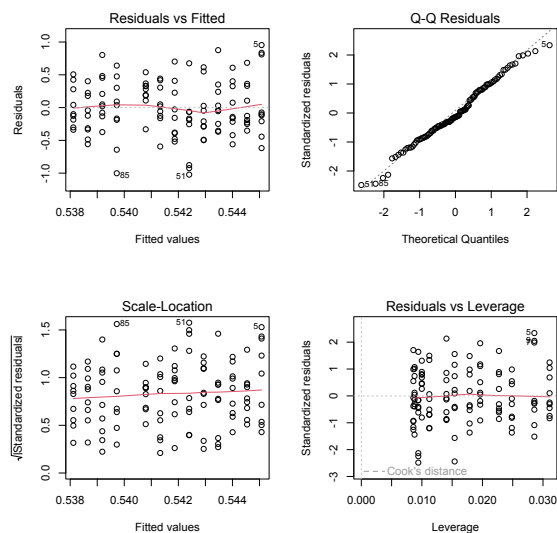
REGRESSION DIAGNOSTIC PLOTS FOR LAKE MICHIGAN

Each set of four plots checks whether the assumptions for linear regression are reasonable after log transformation. The residuals are mostly centered around zero with no clear patterns, which supports a linear relationship. The Q-Q plots show that residuals are close to normally distributed, with only small deviations at the extremes. The scale-location plots suggest that variability is fairly constant, though PTN and POC show slightly more spread at some fitted values. The residuals vs. leverage plots indicate that no single observation has an unusually large influence on the models. Overall, these diagnostics suggest that the regression assumptions are largely met for all four nutrients.

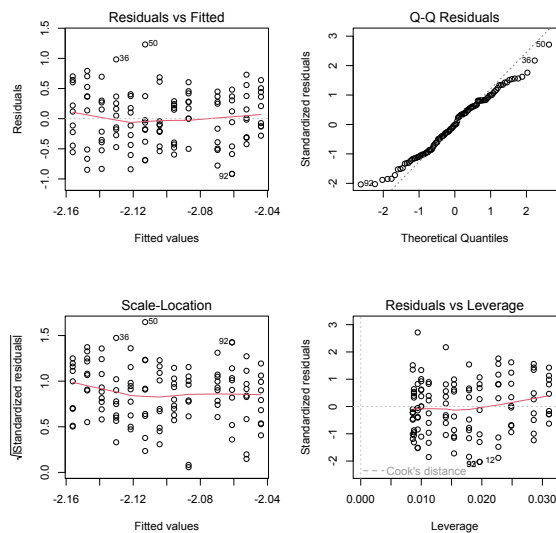
PTN



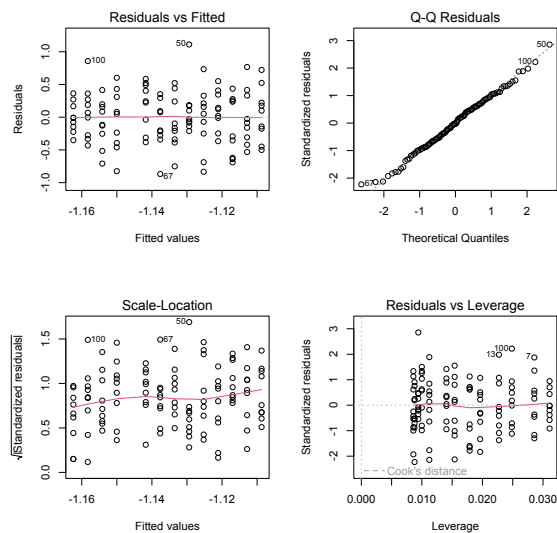
PTP



POC



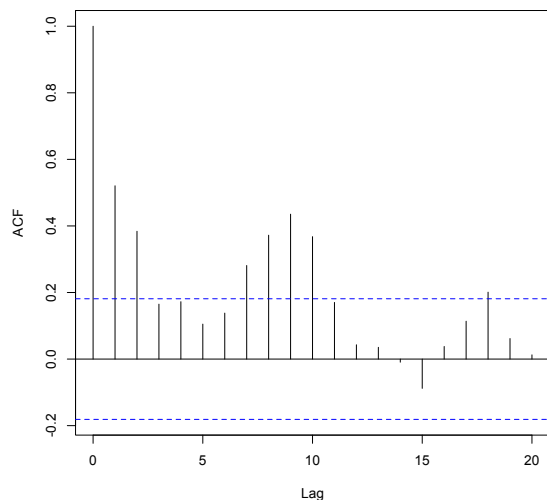
TSS



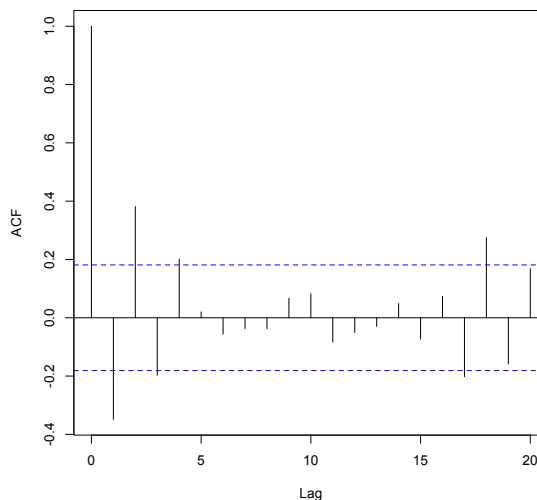
ACF OF REGRESSION RESIDUALS FOR PARTICULATE NUTRIENTS IN LAKE MICHIGAN

These autocorrelation function (ACF) plots show residuals from the log-linear regression models for particulate total nitrogen (PTN), particulate total phosphorus (PTP), particulate organic carbon (POC), and total suspended solids (TSS) over time. The dashed blue lines indicate approximate 95% confidence bounds; spikes extending beyond these bounds suggest residual autocorrelation and potential violations of the independence assumption. PTN and POC display several lags with significant autocorrelation, indicating possible temporal dependence in the residuals. In contrast, PTP and TSS show little to no systematic autocorrelation, suggesting that the independence assumption is reasonably satisfied for those models.

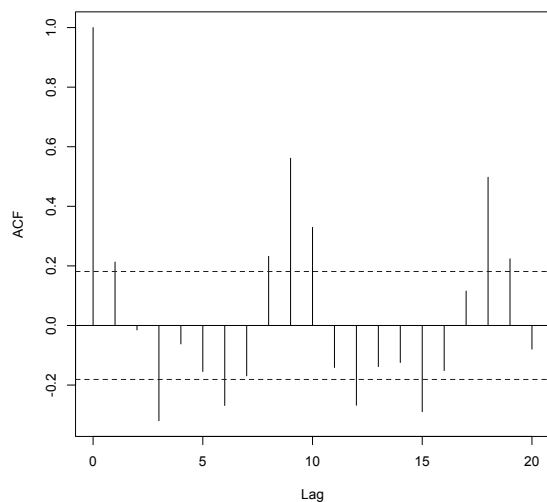
ACF residuals: PTN



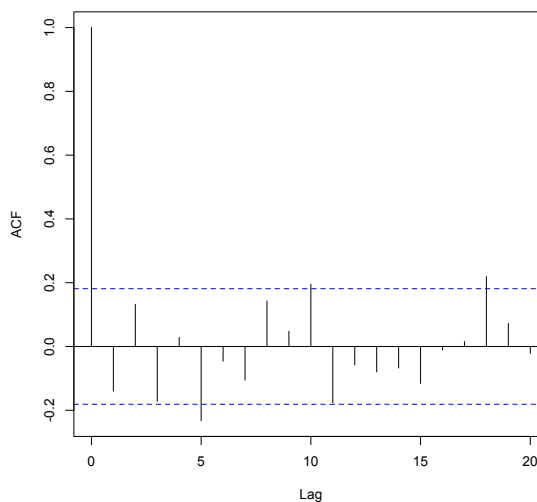
ACF residuals: PTP



ACF residuals: POC



ACF residuals: TSS



QUESTION 2

Can I identify which lake a particulate nutrient sample came from using a combination of particulate nutrient information and seasonal data?

H0: A model using particulate nutrients and season predicts LAKE no better than a baseline model.

H1: A model using particulate nutrients and season predicts LAKE better than the baseline model.

To assess whether particulate nutrient concentrations and season can be used to identify which lake a sample came from, I compared three classification approaches: a baseline model that always predicted the most common lake, a season-only model, and a full model using season plus four particulate nutrients PTN, PTP, POC, and TSS. The data were randomly split into training ($\approx 70\%$) and test ($\approx 30\%$) sets. Model performance was evaluated using test-set accuracy and a confusion matrix computed only on the held-out test data ($n = 150$) to provide an unbiased estimate of out-of-sample performance.

Several assumptions were assessed using visual diagnostics. Performance was evaluated exclusively on a held-out test set, supporting independence and generalization. A comparison of model accuracies shows that improvements are driven by nutrient information rather than season alone. Boxplots of log-transformed particulate nutrients by lake and season show meaningful differences across lakes with some overlap, explaining why classification is imperfect but informative. Statistical significance was assessed using a permutation test, which is appropriate because the hypothesis concerns classification accuracy rather than differences in means.

OUTCOME:

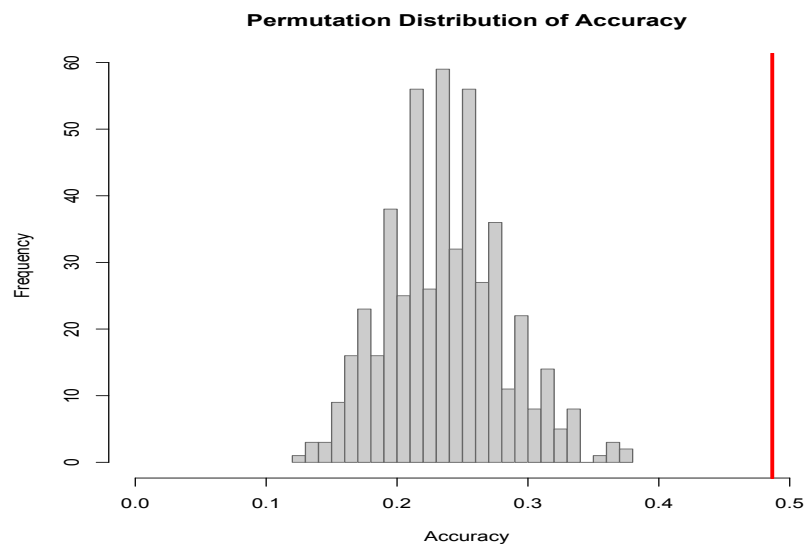
Baseline classification accuracy was ~ 0.23 . Season-only classification accuracy was ~ 0.22 . **Full model classification accuracy was ~ 0.49** , more than twice the baseline accuracy, demonstrating that particulate nutrient measurements contain substantial information about lake identity. The confusion matrix below shows that Lake Erie is classified most accurately, while Lake Ontario is the most difficult to classify. Most misclassifications occur among Lakes Huron, Michigan, and Superior, suggesting overlapping particulate nutrient profiles among these lakes during Spring and Summer.

CONCLUSION:

We can reject H0. The permutation test yielded a p-value of 0.001 (with $B=1000$ permutations), which is actually the lowest possible value we could get since the observed test-set accuracy was higher than every single one of the accuracies we calculated by randomly shuffling the lake labels. This provides really strong evidence that our model, which incorporates particulate nutrients and season, is actually capturing a real signal and predicting lake identity significantly better than just random chance.

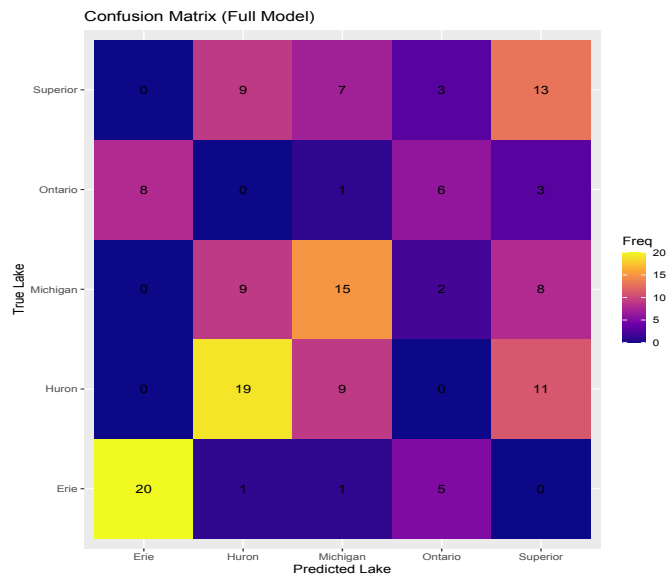
VISUALS:

PERMUTATION DISTRIBUTION ACCURACY



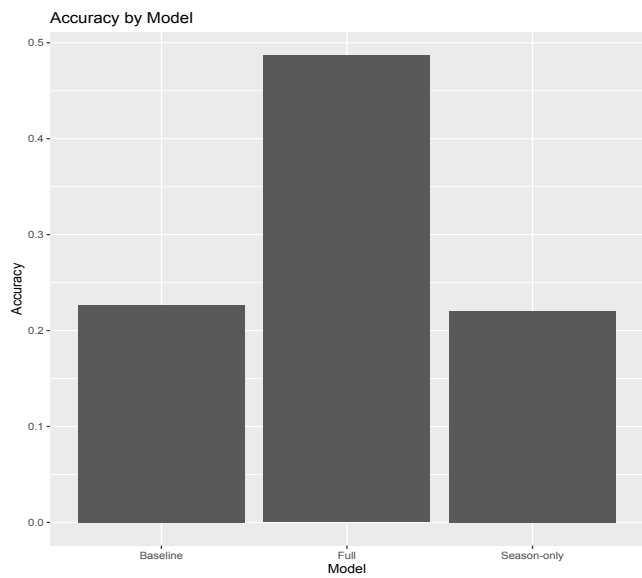
The observed model accuracy of ~48% was outside the range of the permutation distribution, so the x-axis had to be expanded to display it (see the red line).

CONFUSION MATRIX



Confusion matrix showing predicted versus true lake labels for the test set (n = 150). Correct classifications appear along the diagonal (upper right to bottom left corners). Lake Erie is classified most accurately, while most misclassifications occur among Lakes Huron, Michigan, and Superior, indicating overlapping particulate nutrient profiles.

ACCURACY BY MODEL

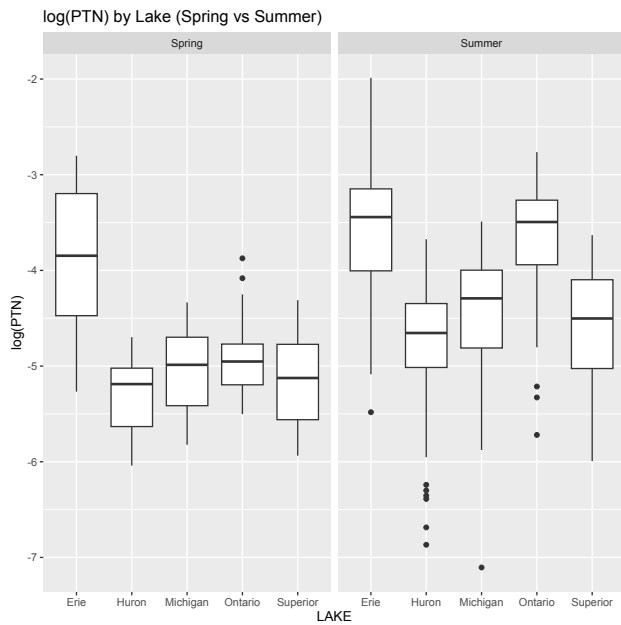


Test-set accuracy for the baseline, season-only, and full models. The full model substantially outperforms both the baseline and season-only models, demonstrating that particulate nutrient variables add meaningful predictive information beyond season.

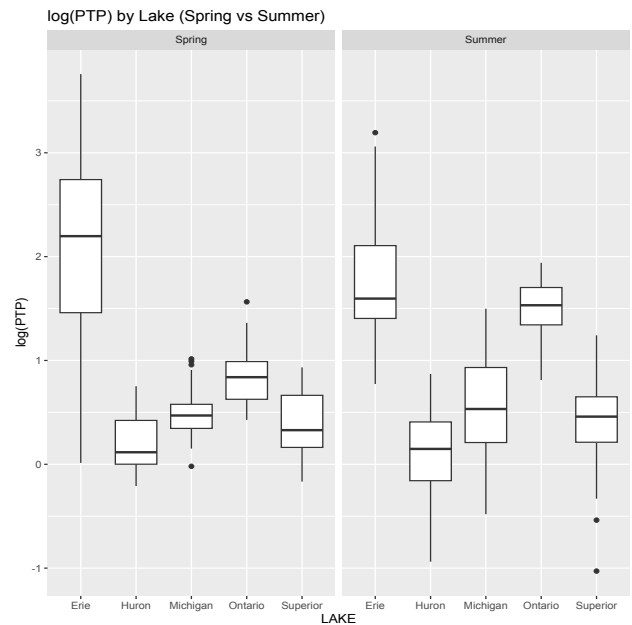
DISTRIBUTION OF LOG() by LAKE (SPRING vs SUMMER) BOXPLOTS

Boxplots of log-transformed particulate nutrients by lake, shown separately for Spring and Summer. Nutrient distributions differ across lakes but overlap substantially, supporting the presence of real signal while explaining imperfect classification performance.

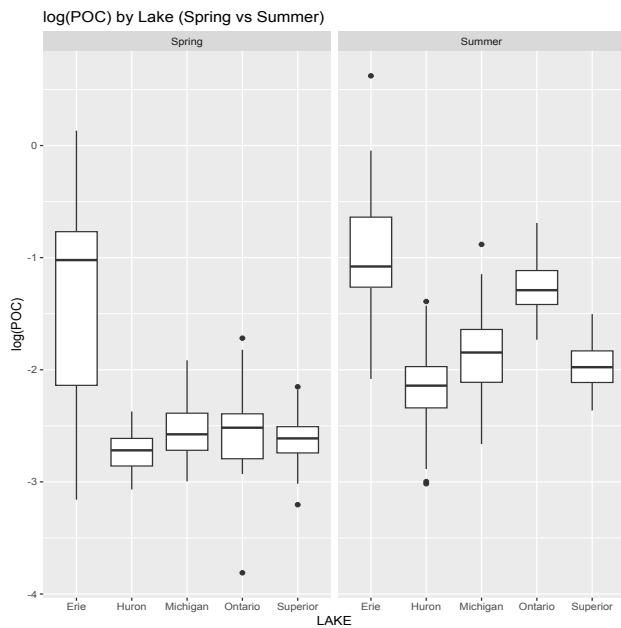
PTN



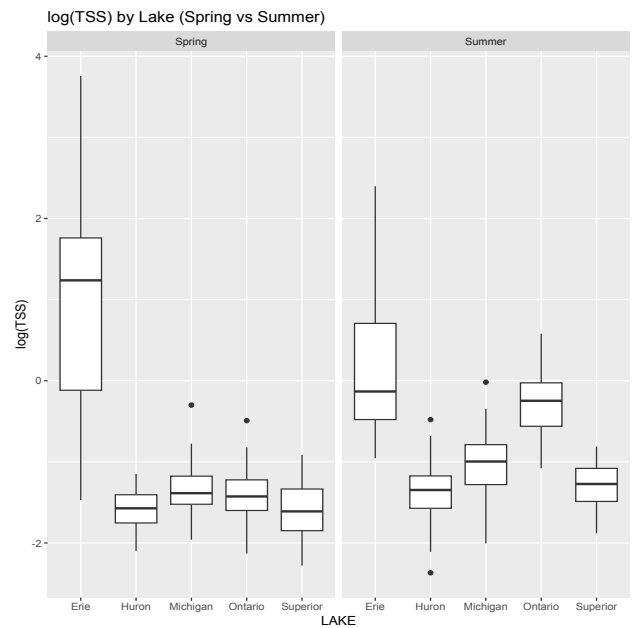
PTP



POC



TSS



QUESTION 3

Are there any outliers in the observations that suggest atypical environmental conditions for any of the lakes?

**** Atypical environmental conditions could be directly related to the season. Hence the hypothesis. ****

H0: After accounting for lake, season does NOT predict outlier status.

H1: After accounting for lake, season DOES predict outlier status.

The purpose of this analysis was to determine whether any observations represent atypical environmental conditions across the Great Lakes and whether these conditions are associated with seasonality. Because unusual lake conditions are often driven by seasonal processes such as runoff, biological activity, or mixing, season was evaluated as a potential predictor of outlier occurrence.

Rather than focusing on average concentrations, this analysis examined extreme values, which are more likely to reflect unusual environmental events. Outliers for four particulate variables—PTN, PTP, POC, and TSS—were identified within each lake using the $1.5 \times \text{IQR}$ rule to ensure anomalies reflected departures from a lake's typical conditions rather than differences across lakes.

Outlier status (yes/no) was modeled using logistic regression. For each nutrient, a lake-only model was compared to a lake-plus-season model using likelihood ratio tests to assess whether season improved model fit beyond lake effects. This approach is appropriate for categorical predictors and does not require normality of the original data.

Noticeably high values were not automatically removed. To distinguish measurement error from plausible environmental events, extreme values were evaluated for consistency across related variables using PTN/TSS, PTP/TSS, and POC/TSS ratios. Observations with high TSS and reasonable ratios were retained as likely environmental events, while isolated or inconsistent extremes were flagged and reviewed separately. Analyses were repeated with and without flagged values to confirm that conclusions were not driven by a small number of observations.

Overall, results indicate that seasonal effects differ by particulate variable. PTN outliers show clear seasonal influence, whereas outliers in PTP, POC, and TSS are more strongly associated with lake-specific factors. These findings highlight the importance of direct outlier analysis in environmental studies and suggest that nitrogen cycling may be particularly sensitive to seasonal processes in the Great Lakes.

OUTCOME:

For each particulate nutrient, the likelihood ratio test comparing Model 1 and Model 2 provided a single p-value used to determine whether to reject or fail to reject the null hypothesis:

- **PTN:**

Reject H_0 : The model comparison returned $p = 0.014$, indicating that adding season significantly improved model fit after accounting for lake. This suggests that PTN outliers are not evenly distributed across seasons and that certain seasons are more associated with extreme PTN values.

- **PTP:**

Fail to Reject H_0 : The likelihood ratio test returned $p = 0.495$, providing no evidence that season improves the model beyond lake. Indicating no detectable seasonal pattern in PTP outliers.

- **POC:**

Fail to Reject H_0 : The likelihood ratio test returned $p = 0.424$, similarly showing no significant contribution of season after accounting for lake.

- **TSS:**

Fail to Reject H_0 : The likelihood ratio test returned $p = 0.396$, again indicating no evidence of a seasonal effect on outlier status once lake differences were considered.

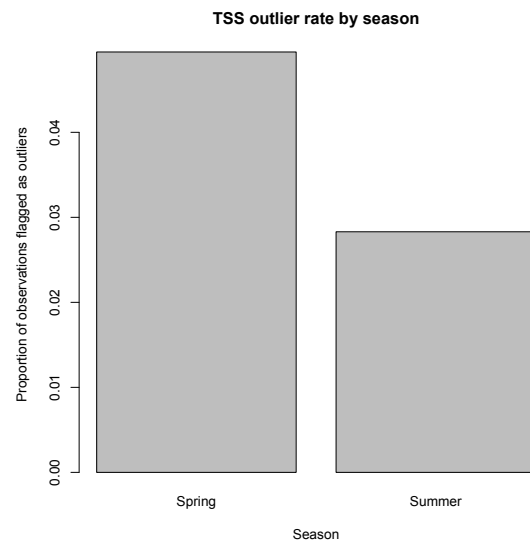
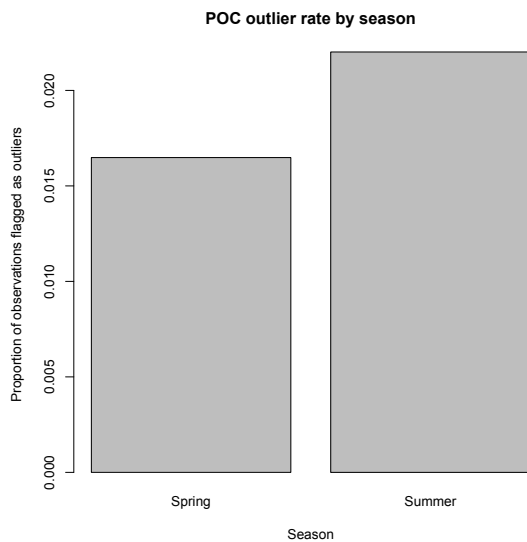
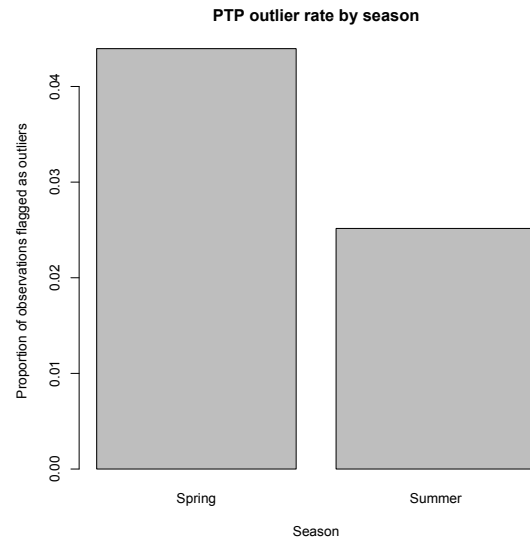
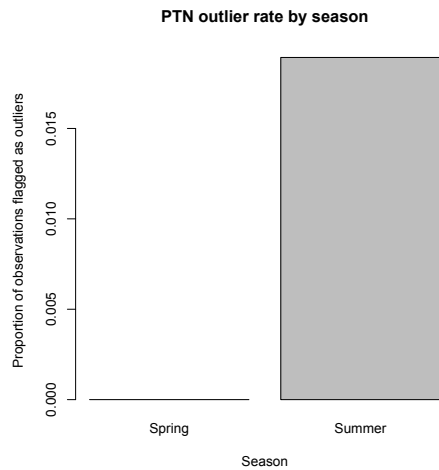
CONCLUSION:

Taken together, this analysis provides evidence that **seasonal effects on anomalous conditions are nutrient-specific rather than universal**. Among the four particulate measures examined, **only PTN showed a statistically significant association between season and outlier occurrence after controlling for lake**. In contrast, PTP, POC, and TSS exhibited outliers that appear to be driven primarily by lake-specific factors rather than seasonal patterns. These findings suggest that seasonal processes may play a role in extreme nitrogen dynamics, while anomalies in phosphorus, organic carbon, and suspended solids are more likely attributable to local or lake-specific conditions.

VISUALS:

SEASONAL DIFFERENCES IN OUTLIER RATES FOR PARTICULATE NUTRIENTS

These bar plots show the **proportion** of observations flagged as outliers (using the $1.5 \times \text{IQR}$ rule within each lake) for particulate total nitrogen (PTN), particulate total phosphorus (PTP), particulate organic carbon (POC), and total suspended solids (TSS), separated by season (spring vs. summer). The visuals are used to assess whether outliers occur more frequently in one season, which directly supports the hypothesis test evaluating whether season predicts outlier status after accounting for lake. Overall, some nutrients show higher outlier rates in spring (notably PTP and TSS), while others show higher rates in summer (PTN and POC), suggesting that seasonal processes may contribute to atypical observations rather than outliers being purely random or due to measurement error.

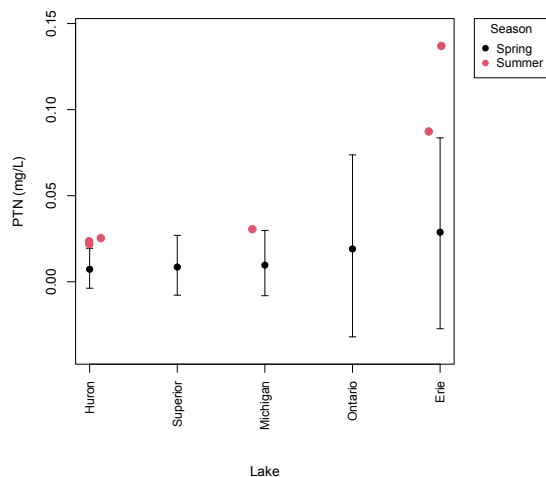


OUTLIER DISTRIBUTIONS FOR PARTICULATE NUTRIENTS BY LAKE AND SEASON

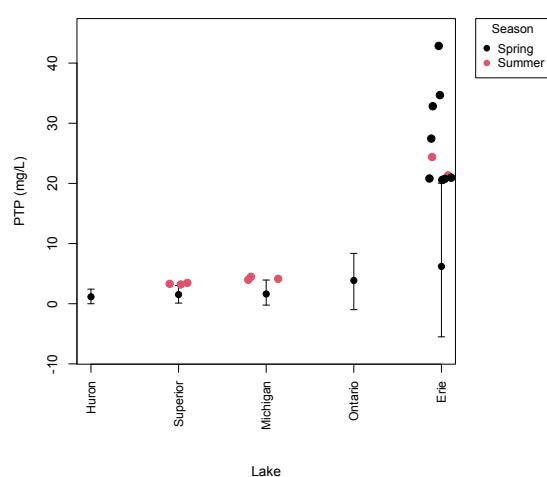
Outliers for particulate total nitrogen (PTN), particulate total phosphorus (PTP), particulate organic carbon (POC), and total suspended solids (TSS) were identified using the standard $1.5 \times \text{IQR}$ rule, calculated separately for each lake. Defining outliers within each lake helps ensure that extreme values reflect unusual conditions relative to that lake, rather than differences in typical nutrient levels across lakes. The points shown represent observations classified as outliers, with color indicating season (spring or summer). The error bars show the interquartile range for each lake–nutrient combination. Together, these plots make it easier to compare how large the extreme values are and whether outliers tend to occur more often in a particular season across different lakes.

*As an aside, I'm really interested in what's driving Lake Erie's extreme particulate measures.

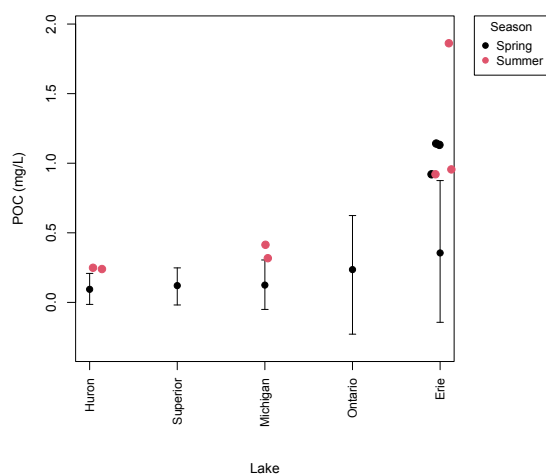
PTN outlier bounds by lake (IQR) with outliers by season



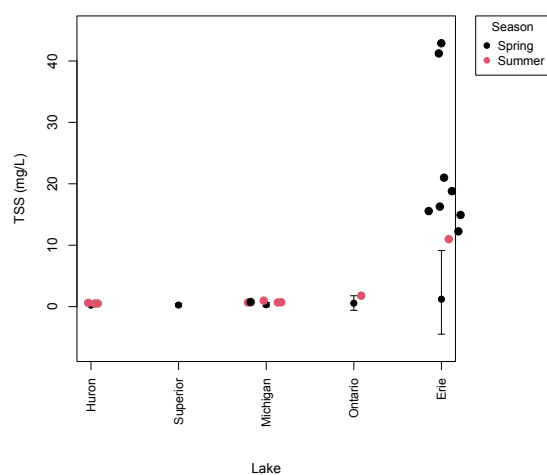
PTP outlier bounds by lake (IQR) with outliers by season



POC outlier bounds by lake (IQR) with outliers by season



TSS outlier bounds by lake (IQR) with outliers by season



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